

Table S13. Chronotype and timing

Metric	Instrument	Signed association, h (95% CI)	FDR-adjusted p	Fitted sample: participants / days / observations / hours / sites
Midpoint of the brightest 10 hours	MCTQ MSFsc	+0.205 (+0.045 to +0.366)	0.017	140 / 810 / 810 / 18711.7 h / 9
Midpoint of the brightest 10 hours	MEQ	-0.278 (-0.451 to -0.105)	0.002	141 / 816 / 816 / 18851.0 h / 9
Midpoint of the darkest 10 hours	MCTQ MSFsc	+0.276 (+0.118 to +0.435)	0.003	140 / 810 / 810 / 18711.7 h / 9
Midpoint of the darkest 10 hours	MEQ	-0.314 (-0.487 to -0.142)	0.001	141 / 816 / 816 / 18851.0 h / 9
First light timing above 250 lx melEDI	MCTQ MSFsc	+0.381 (+0.146 to +0.616)	0.003	139 / 722 / 722 / 16716.3 h / 9
First light timing above 250 lx melEDI	MEQ	-0.444 (-0.699 to -0.189)	0.001	140 / 727 / 727 / 16832.5 h / 9
Last light timing above 250 lx melEDI	MCTQ MSFsc	-0.025 (-0.264 to +0.214)	0.811	140 / 683 / 683 / 15900.8 h / 9
Last light timing above 250 lx melEDI	MEQ	+0.029 (-0.236 to +0.294)	0.814	141 / 687 / 687 / 15995.0 h / 9
Midpoint of the longest continuous period above 250 lx melEDI	MCTQ MSFsc	+0.161 (-0.113 to +0.434)	0.284	131 / 478 / 478 / 11325.5 h / 9
Midpoint of the longest continuous period above 250 lx melEDI	MEQ	-0.228 (-0.521 to +0.064)	0.135	132 / 482 / 482 / 11419.8 h / 9

MCTQ associations are per one-hour later MSFsc; MEQ associations are per 10 points greater morning preference. Positive associations indicate later timing and negative associations earlier. Intervals are 95% Wald CIs. FDR-adjusted p-values are bold when FDR-adjusted p < 0.050 within the instrument-specific five-outcome family.